2AMM15 : Final Assignment

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April 2, 2026

Introduction

This report outlines the design decisions, results and AI usage during my take on the final assignment. The report is split up per part corresponding to the Jupyter Notebook. I have made this assignment on my personal computer, as it has two GPUs available for training. I had to reinstall torch to support CUDA on Windows however. I first tried this myself, but used GPT5-mini to troubleshoot once I ran into errors. It did work fine on my Ubuntu installation.

1 Part 1

This section goes over data augmentation, training the ResNet-18 and ViT-Base models, the issues encountered while training and a comparison between the two models.

1.1 Approach and challenges encountered

In order for the models to be more robust against real-world noise, I augmented the dataset by flipping each image horizontally with a probability of 50%, and a random rotation of at most 15 degrees. This because images can be taken from essentially random angles, and our models should be robust enough to still correctly identify the skin conditions.
Loading the pre-trained ResNet-18 model was straightforward, as well as changing the final fully-connected layer. I started out with a rather standard train_model function, with the model, train dataset loader, test dataset loader, device, number of epochs, learning rate and weight decay as parameters. While training the first ResNet-18 model, I noticed that the validation accuracy flattened out pretty quickly, meaning it stopped improving marginally after the first few epochs already. This led me to implementing an early stopping patience, that is, the amount of epochs it would continue for when no loss improvement was seen, to prevent overfitting and computation wastage. Given the default configuration ran for a total of 20 epochs, I thought an early stopping patience of 3 was good to start with. However, when training ResNet-18 for the first time with *early_stopping_patience* set to 3, it stopped at epoch 6 (counting from 0). However, when looking at the loss and accuracy curves, it stopped at an epoch which significantly increased the loss (and decreased accuracy), see figure 1. Hence, I tried 3 different *early_stopping_patience* values; 3, 6 and 100. Since 100 is much bigger than the amount of epochs, there was no early stopping during this configuration. A value of 100 is also called *no_patience* throughout the notebook. Training time per epoch was around 20 seconds for all ResNet-18 configurations.
The ViT-Base model was, as with the ResNet-18 model, straightforward to load and modify, replacing it's default classification head to one with 7 output classes instead of 1000. To train the ViT-Base model, I opted for a slower learning rate (half of ResNet-18's learning rate, at 0.0005), and tested 3 different early stopping patience values, 3, 6 and 100 (or no patience). I initially ran the code for all three configurations in one code cell, but I split this up into 3 after the notebook kept stopping halfway through the third configuration training. Doing it one cell at the time worked. Training time per epoch was around 1 minute and 30 seconds for all ViT-Base configurations.

1.2 Comparison

We will first go over the ResNet-18 model by itself. As seen in figure 1, but also in figures 2 and 3, we see that even though the training loss goes down and the training accuracy goes up, both validation loss and accuracy fluctuate, no matter the early stopping patience value. The model is overfitting and this is most likely due to the learning rate being too high for a pre-trained model being finetuned on this data. At *early_stopping_patience* 3 and 6 however, the model was saved when its loss was minimized and accuracy maximized, meaning our best ResNet-18 model got a validation accuracy of over 80%. I was, due to time, not available to check whether it was actually the learning rate. As seen in figures 4, 5 and 6, we do not see see the validation loss and validation accuracy fluctuating, but rather (as expected) shrinking and growing together with their training variant. As we can see, only for *early_stopping_patience* 3 was the ViT-Base model saved after 17 epochs. For the other configurations, it sat through the entire 20 epochs of training.
Based on validation accuracy, the two best models of all configurations were ResNet-18 with no patience, at 82.75% accuracy and ViT-Base with a patience of 6 at 76.37% accuracy. These values can be derived from the comparison table found at the end of Part 1 in the Jupyter Notebook.
We will now go over all critical classifications, such as Melanoma and basal cell carcinoma. The Resnet-18 model classified Melanoma right only 88 times of the 223 Melanoma cases (39,46% true prediction rate). 105 out of 223 cases were classified as Melanocytic Nevi, which are common moles, but look similar to Melanoma. Basal cell carcinoma had a much higher true prediction rate of 79,61% (82 out of 103 true cases), with it being mostly misclassified as melanocytic nevi.
The ViT-Base model classified Melanoma right only 48 times out of 223 cases (21,52% true prediction rate). It misclassified 136 Melanoma instances as melanocytic nevi. Basal cell carcinoma had a much higher true prediction rate of 60,19% (62 out of 103 true cases), with it being mostly misclassified as melanocytic nevi.
This leads to a clear winner in on this dataset, namely the ResNet-18 model. Not only does it train faster and have a higher overall accuracy, it correctly classifies critical skin conditions much better than the ViT-Base model. What the ViT-Base model does a little bit better, is correctly classifying melanocytic nevi. This is probably due to the amount of melanocytic nevi images in the training set, as it is overrepresented in the dataset, which ViT-Base is susceptible for.

2 Part 2

Figures 9, 11, 13, 15, 17 represent the gradient visualizations of the final convolution layers in ResNet-18 on the 5 images chosen in step 1. Figures 10, 12, 14, 16, 18 represent the self-attention weights across multiple heads and layers of the Vision Transformer on the 5 images chosen in step 1.
On image A, where both models classified the lesion correctly with a high confidence, we see that the patches of the ViT model are much more centered, covering all of the lesion as seen in figure 10, while the ResNet-18 model is focusing more on the top of part of the lesion as seen in figure 9. There is a small white hair visible in this image, which might have thrown off the ResNet-18 model a bit when looking for structures within the image.
On image B, where both models classified the lesion correctly but with low confidence, we see that the ResNet-18 model focuses on the top of the lesion again as seen in figure 11, presumably because of the darker spot and possibly the hairs surrounding it. The ViT model seems to capture more of the lesion, but is fragmented, as the red patches are not close together, as seen in 12.
On image C, where the CNN is correct and the ViT is not, we see the CNN focuses on the whole lesion and its hotspot is focused on the center of it (figure 13). While the ViT model does cover the lesion, its red areas are scattered throughout the lesion (figure 14). Due to the local context, like texture and shapes within the lesion, the CNN has the advantage over the ViT model which uses a more global context, which was not enough to correctly classify the lesion.
On image D, where ViT is correct but the CNN is not, we see in figure 15 that the CNN captures the lesion spot on, but fails to classify correctly, probably because benign keratosis-like lesions share local features like textures and color, but only the ViT was able to correctly classify this image due to its global context (16, seeing that the lesion was surrounded by a more pink skin.
On image E, where the image contains an artifact, we have that both models correctly classify the lesion, but both with a low confidence. Looking at their heatmaps in figures 17 and 18, we see that the

CNN focuses on the lesion, but has yellow-green parts scattered all around the center too, the shape of the heatmap does not correspond to the shape of the lesion, seemingly putting more emphasis on the artifacts. The ViT model's attention is scattered throughout the image, where it looks like it is distracted by the artifacts as well, putting high focus on patches with high contrast.
When comparing implementations regarding their applicability in a healthcare situation, I assume the GradCAM implementation would be of more help than the attention rollout implementation, as the GradCAM implementation is easier to comprehend, allowing users to know what the model actually focused on to get to its decision, while this is not always clear for the attention rollout implementation, due to the possibility of scattered red zones across the image.

3 Part 3

For the third part, 3 configurations have been set up for fine-tuning:

1. Configuration 1: r = 16, alpha = 32, dropout = 0.05, LR = $5 * 10^{-5}$, on all 7 modules for 2 epochs.

2. Configuration 2: r = 32, alpha = 64, dropout= 0.05, LR = $2 * 10^{-4}$, on all 7 modules for 3 epochs.

3. Configuration 3: r = 16, alpha = 32, dropout = 0.1, LR = $2*10^{-4}$, on q/k/v/o only for 3 epochs.

The first configuration was my initial configuration, with a smaller learning rate and just two epochs to prevent overfitting. The second configuration doubles the rank, meaning each module has twice as many parameters to represent the weight update, allowing the model to better learn the details within the training data. Alpha is scaled proportionally. The third configuration removes the MLP projections from adapting, only allowing the attention projections to be trained. The dropout here is higher to prevent overfitting due to the smaller adapter set.
The first configuration had its evaluation loss at 0.340108, the second at 0.341125, and the third on 0.339800. It turns out that adapting only q/k/v/o was sufficient to get a better model. Training configuration three also took the least amount of time, 32 minutes as opposed to 41 and 47 minutes for configuration 1 and 2 respectively.
The baseline model only had an accuracy of 25%. After fine-tuning using configuration C, we achieved an accuracy of 31%, which is a 6% increase. This however is still not a good accuracy, certainly not for the use case in healthcare.

4 Figures

4.1 Part 1

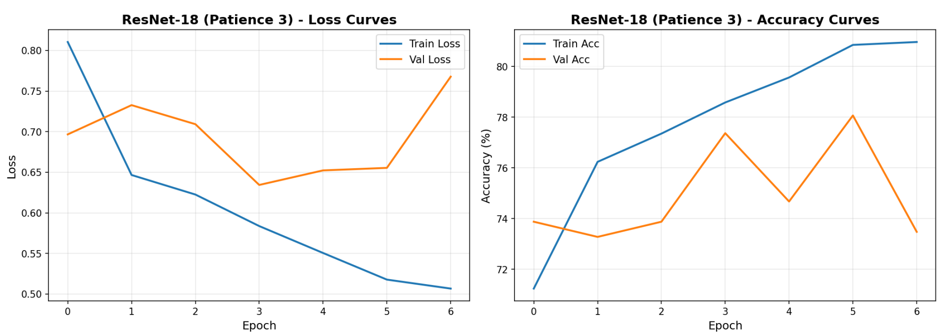

Figure 1: Training history of ResNet-18 with patience set to 3.

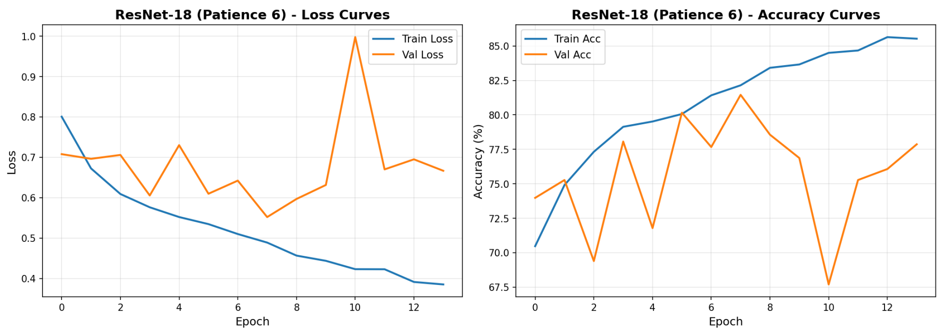

Figure 2: Training history of ResNet-18 with patience set to 6.

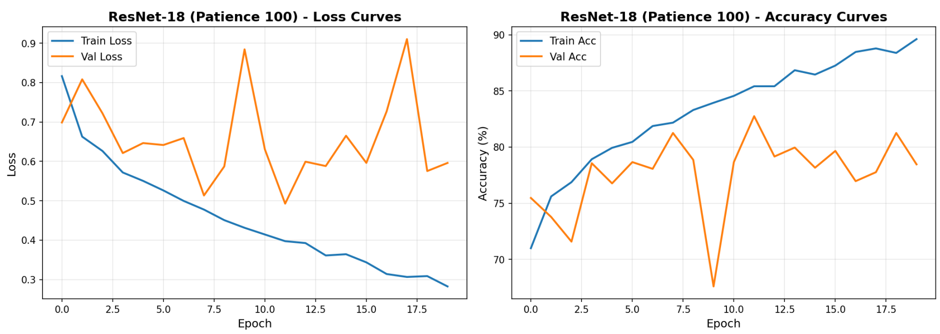

Figure 3: Training history of ResNet-18 with patience set to 100.

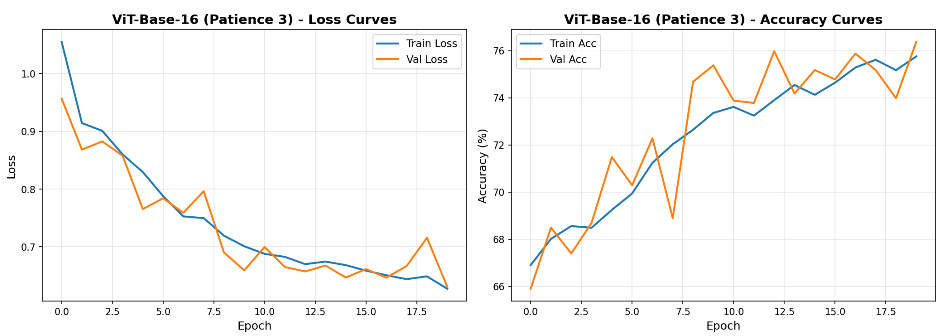

Figure 4: Training history of ViT-Base with patience set to 3.

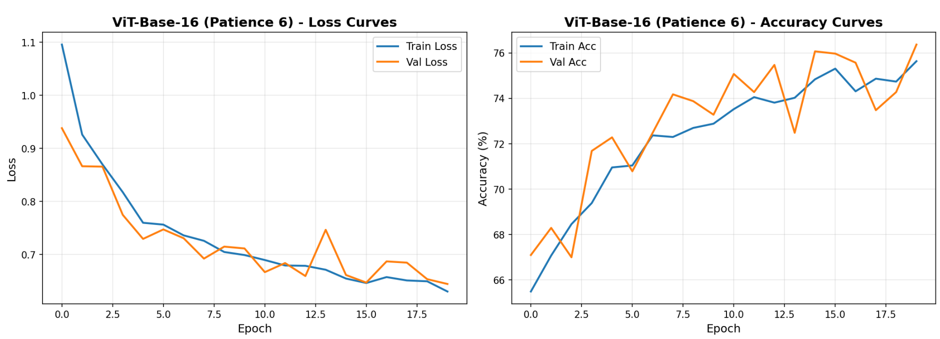

Figure 5: Training history of ViT-Base with patience set to 6.

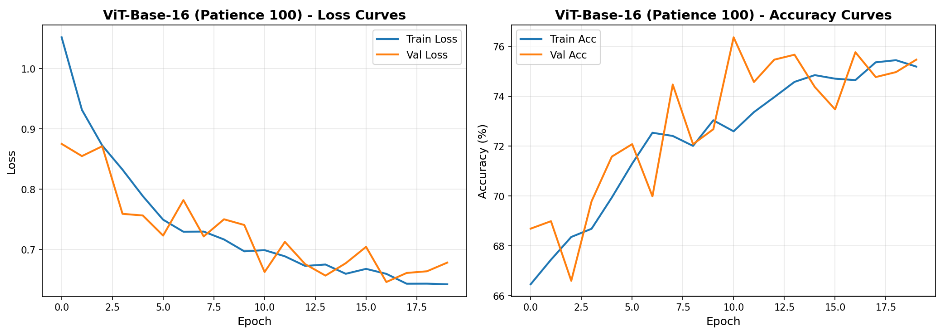

Figure 6: Training history of ViT-Base with patience set to 100.

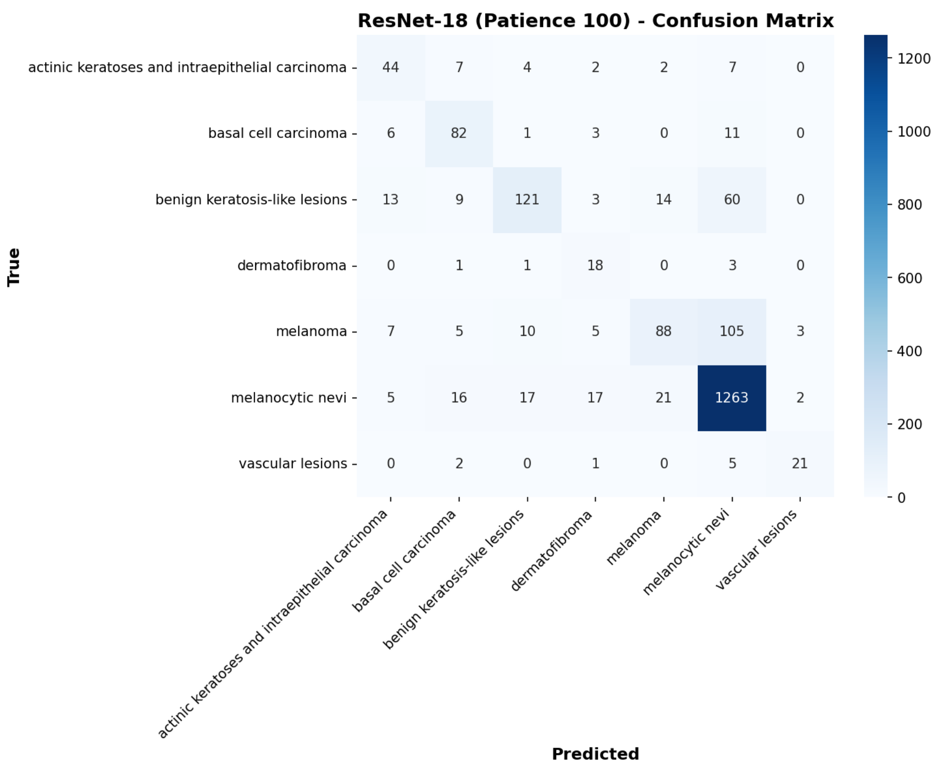

Figure 7: Confusion matrix of ResNet-18 with patience set to 100.

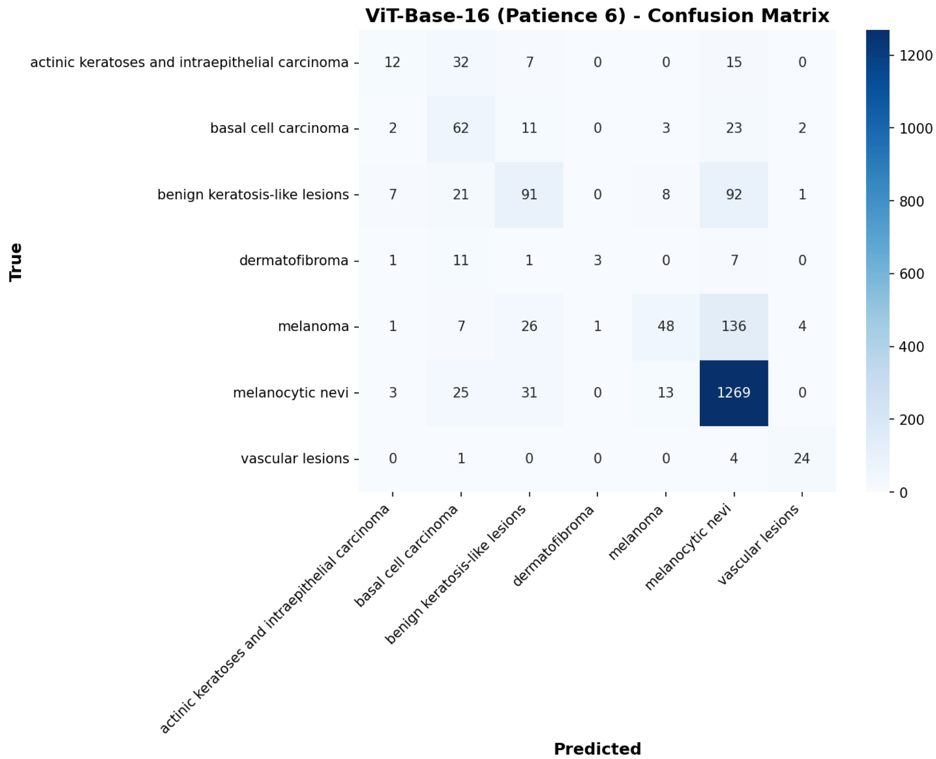

Figure 8: Confusion matrix of ViT-Base with patience set to 6.

4.2 Part 2

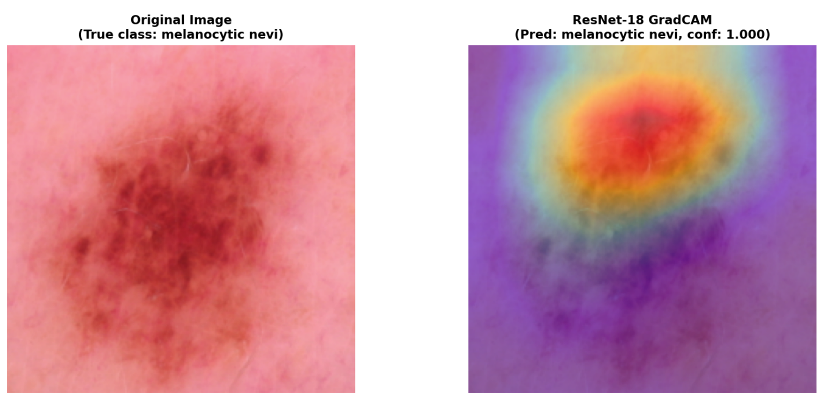

Figure 9: Grad-CAM visualization for ResNet-18 on image A.

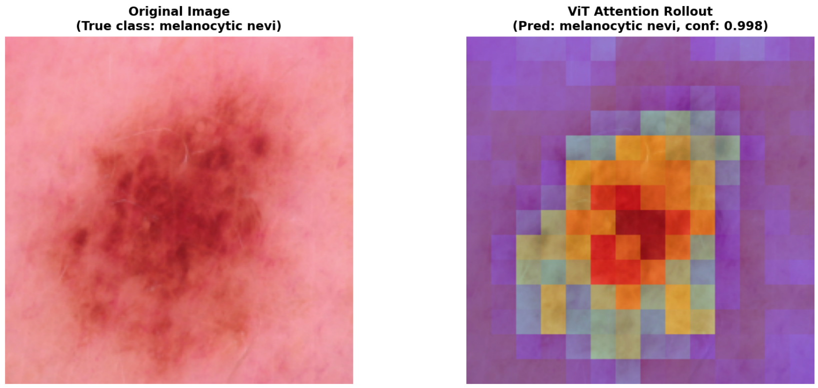

Figure 10: ViT rollout visualization for image A.

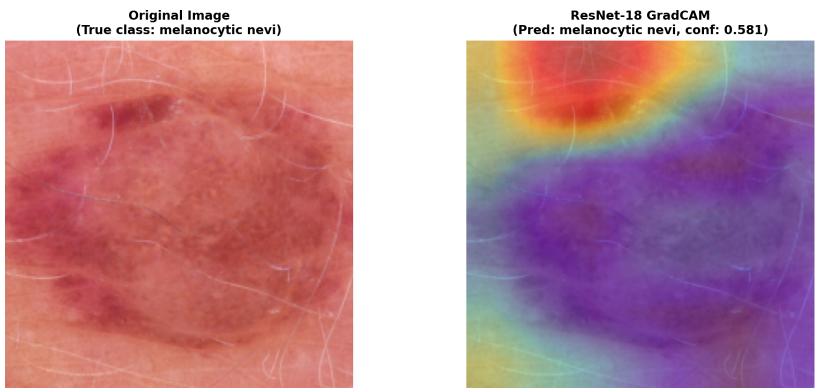

Figure 11: Grad-CAM visualization for ResNet-18 on image B.

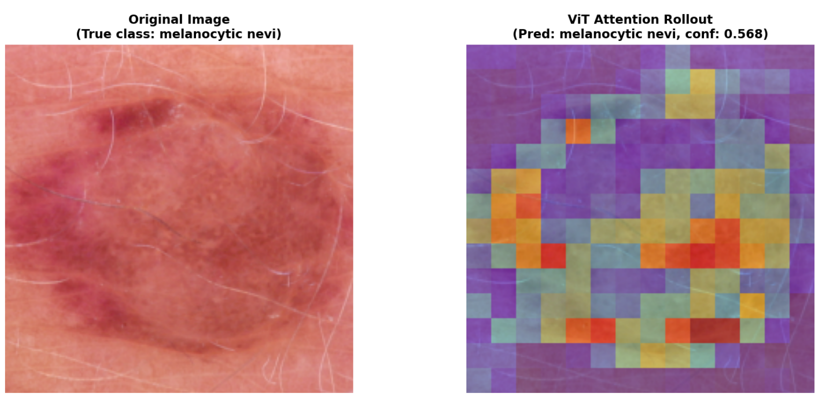

Figure 12: ViT rollout visualization for image B.

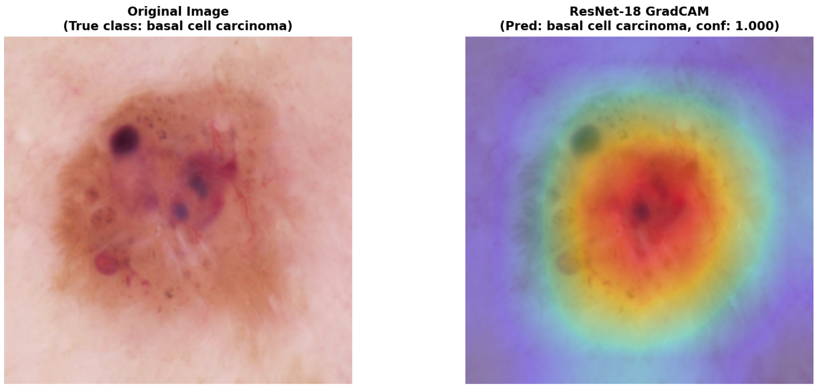

Figure 13: Grad-CAM visualization for ResNet-18 on image C.

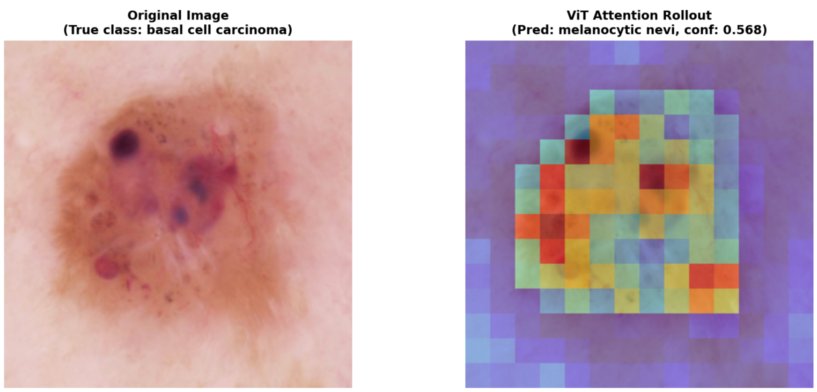

Figure 14: ViT rollout visualization for image C.

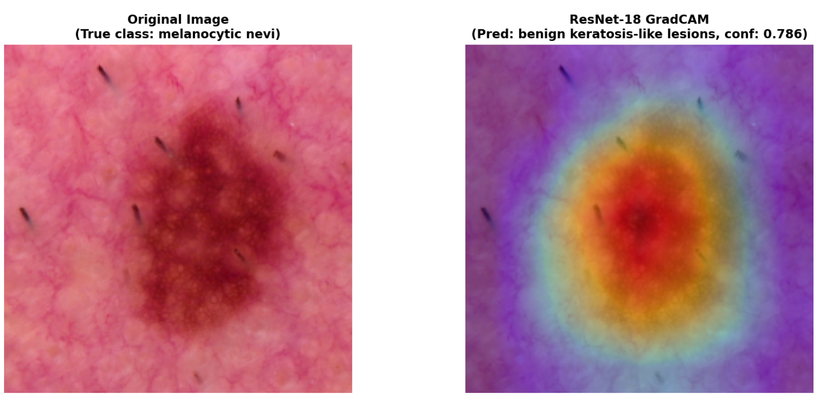

Figure 15: Grad-CAM visualization for ResNet-18 on image D.

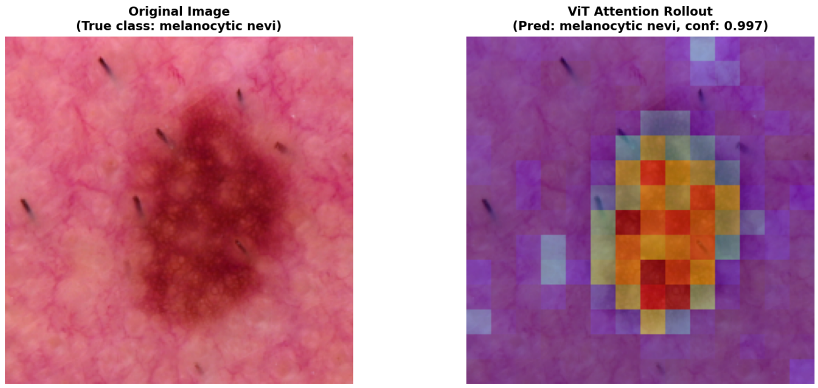

Figure 16: ViT rollout visualization for image D.

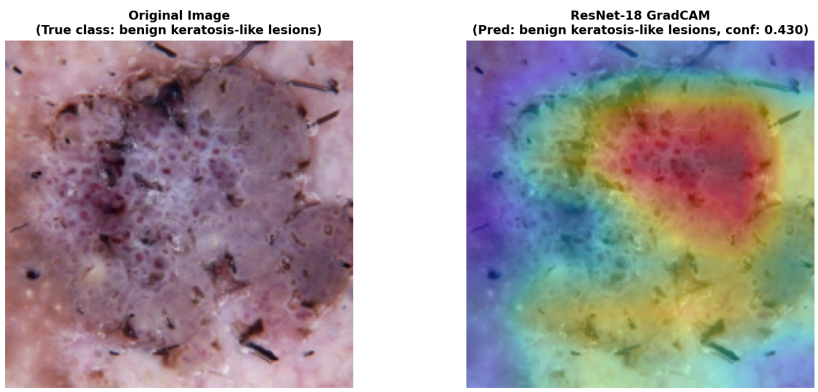

Figure 17: Grad-CAM visualization for ResNet-18 on image E.

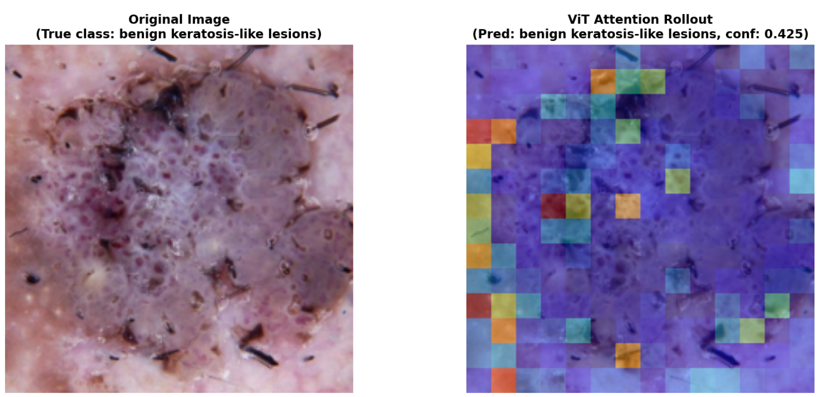

Figure 18: ViT rollout visualization for image E.